

HYBRID RAG FRAMEWORK FOR ENTERPRISE KNOWLEDGE MANAGEMENT

TEAM MEMBERS

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INTRODUCTION

Modern organizations generate vast amounts of enterprise documents such as annual reports, company policies, technical manuals, financial statements, product documentation, and knowledge base articles. As these documents continue to grow, finding accurate information becomes difficult using traditional keyword-based search systems.

Large Language Models (LLMs) provide intelligent responses but often generate incorrect or hallucinated answers when they lack access to enterprise-specific information.

Retrieval-Augmented Generation (RAG) improves this by retrieving relevant enterprise documents before generating responses. This project further enhances RAG using a **Hybrid Retrieval** approach that combines semantic and keyword search for more accurate and context-aware enterprise knowledge management.

PROBLEM STATEMENT

Organizations face several challenges while managing enterprise knowledge:

- Large volumes of enterprise documents
- Slow manual document search
- Traditional keyword search misses contextual information
- Semantic search alone may miss exact keyword matches
- LLMs produce hallucinated responses without enterprise context
- Employees spend significant time searching for information

There is a need for an intelligent enterprise assistant capable of retrieving accurate information from organizational documents.

OBJECTIVES

The objectives of the proposed system are:

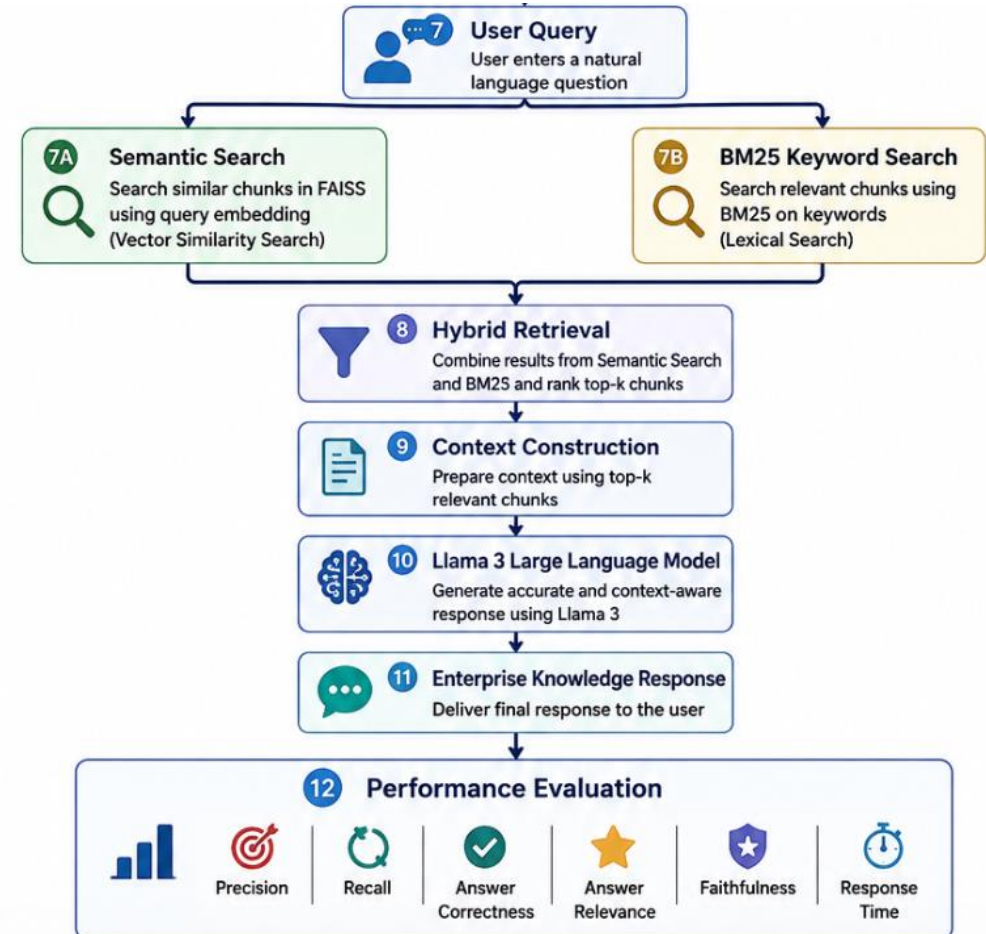
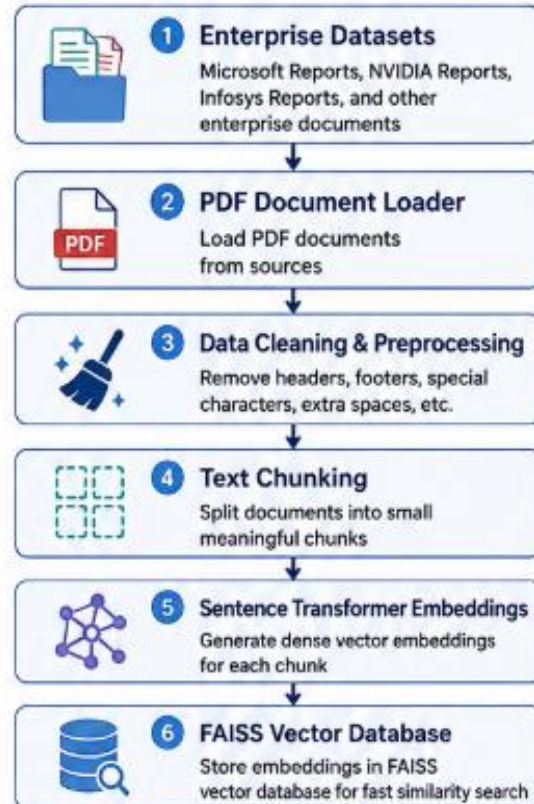
- **Improve retrieval accuracy** by overcoming the limitations of semantic-only retrieval.
 - **Combine semantic and keyword-based search** using FAISS and BM25.
 - **Improve relevance of retrieved context** for enterprise-specific queries.
 - **Reduce irrelevant or incomplete responses** caused by ineffective document retrieval.
 - **Reduce LLM hallucinations** by providing more relevant and reliable context.
 - **Evaluate the proposed Hybrid RAG against the existing RAG approach** using multiple enterprise datasets.
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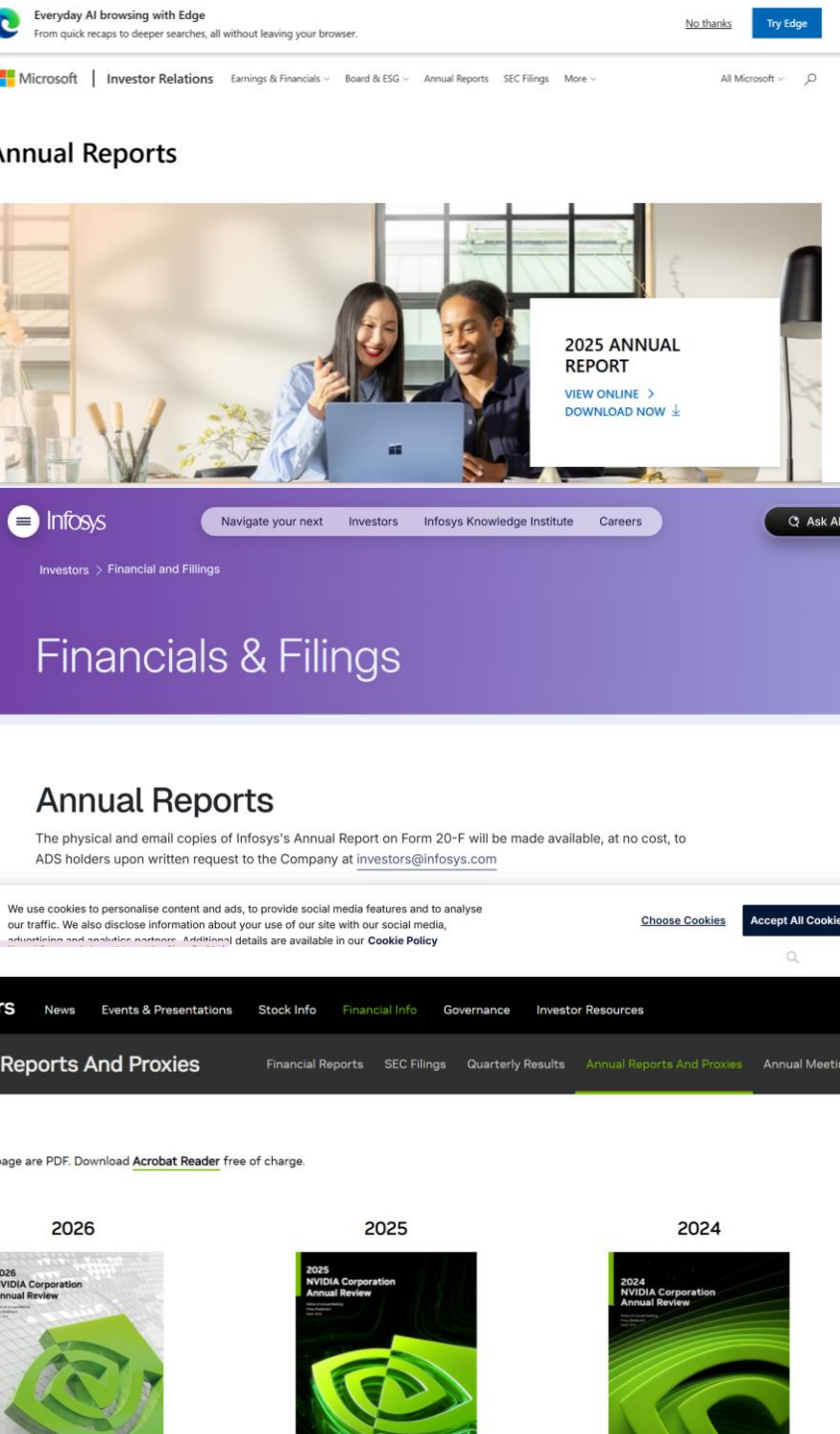
LITERATURE SURVEY

S.No	Existing Project	What They Did	Limitation	Our Improvement
1	EKRAG: Benchmark RAG for Enterprise Knowledge QA (2025)	Enterprise question answering using RAG and enterprise documents.	Uses semantic retrieval only.	Hybrid Retrieval (Semantic + BM25) for improved accuracy.
2	EnterpriseRAG-Bench (2026)	Evaluated RAG systems on enterprise documents from Slack, Jira, GitHub, Gmail, etc.	Focuses on benchmarking, not a complete enterprise assistant.	Build a deployable enterprise knowledge management system.
3	Secure Multifaceted-RAG (2025)	Used RAG with enterprise reports and secure retrieval mechanisms.	High computational complexity and security-focused.	Lightweight Hybrid RAG with faster retrieval and easier deployment.
4	Agentic RAG with Hybrid Retrieval (2026)	Multi-agent RAG with role-based access control for enterprises.	Complex architecture requiring multiple AI agents.	Simpler Hybrid RAG using LangChain, FAISS, BM25, and Llama 3.
5	Enterprise Document Intelligence using RAG	AI assistant for enterprise documents and business reports.	Uses only vector similarity search.	Combines vector search with keyword retrieval for better context retrieval.

FLOW CHART

HYBRID RAG FRAMEWORK FOR ENTERPRISE KNOWLEDGE MANAGEMENT





DATASETS

The proposed system uses **publicly available enterprise documents** from three companies:

- **Microsoft Annual Reports** – Business, financial, and sustainability reports.
- **NVIDIA Annual Reports** – AI, technology, and corporate reports.
- **Infosys Annual Reports** – Annual, ESG, and governance reports.

These datasets are used to evaluate the performance and effectiveness of the proposed Hybrid RAG framework in enterprise knowledge retrieval.

EXPECTED OUTCOME

The proposed **Hybrid RAG Framework for Enterprise Knowledge Management** is expected to provide fast, accurate, and context-aware responses by combining semantic search and keyword based retrieval. The system will efficiently retrieve relevant information from enterprise documents such as annual reports and corporate publications, reducing hallucinations and improving answer relevance. It aims to enhance enterprise knowledge accessibility, decision-making, and overall information retrieval performance.

CONCLUSION

The proposed Hybrid RAG Framework offers an effective solution for enterprise knowledge management by integrating semantic retrieval, BM25 keyword search, and Llama 3. By leveraging publicly available enterprise documents from Microsoft, NVIDIA, and Infosys, the system delivers reliable and context-aware responses. Compared to traditional RAG systems, the proposed framework improves retrieval accuracy, response quality, and user experience, making it a practical solution for modern enterprise information management.

THANK YOU